Exploring Roles and Purposes in More-than-Human Design through a Reflexive Design Studio Experiment

Çağlar Genç
Tampere University
Tampere, Finland
id.caglargenc@gmail.com

Ferran Altarriba Bertran*†
Tampere University
Tampere, Finland
EU ERAM, Universitat de Girona
Girona, Spain
ferran.altarribabertran@tuni.fi

Oğuz Buruk*
Tampere University
Tampere, Finland
oguz.buruk@tuni.fi

Sangwon Jung*
Tampere University
Tampere, Finland
sangwon.jung@tuni.fi

Velvet Spors*
University of Tartu
Tartu, Estonia
velvet.spors@ut.ee

Juho Hamari
Tampere University
Tampere, Finland
juho.hamari@tuni.fi

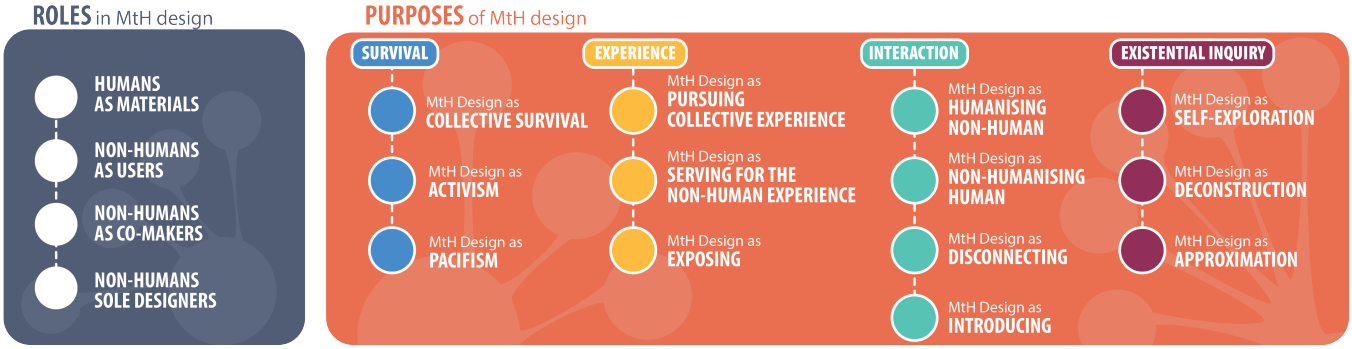

Figure 1: The list of Roles and Purposes in MtH Design that are formulated based on our Reflexive Design Studio.

Abstract

We are witnessing a shift towards a more-than-human (MtH) paradigm in HCI, recognizing non-human beings as interconnected in our existence, from daily life to design. While exciting MtH design works have emerged, they often rely on abstract concepts (e.g., decentering) that remain open to interpretation, leaving practical nuances underexplored. To address this, we adopted a *bottom-up Research through Design* approach, a *Reflexive Design Studio*: as a design team, we created and reflected on conceptual designs using the mushroom basket as a metaphor to explore human–non-human relationalities. From these, we articulated *roles* (e.g., *humans as materials*, *non-humans as co-makers*) and *purposes* (e.g., *activism*, *collective survival*, *approximation*) that illustrate how MtH design might be practiced with nuances. Relating these to theory and prior work, we contribute an exploratory framework offering generative and critical starting points for future MtH design research to test, contest, expand, and engage with MtH complexities in practice.

CCS Concepts

• **Human-centered computing → Human computer interaction (HCI)**; **Collaborative and social computing**.

Keywords

post-humanism, more-than-human, design, entanglement, relational ontologies

ACM Reference Format:
Çağlar Genç, Ferran Altarriba Bertran, Oğuz Buruk, Sangwon Jung, Velvet Spors, and Juho Hamari. 2025. Exploring Roles and Purposes in More-than-Human Design through a Reflexive Design Studio Experiment. In *28th International Academic Mindtrek (Mindtrek '25), October 07–10, 2025, Tampere, Finland*. ACM, New York, NY, USA, 13 pages. https://doi.org/10.1145/3757980.3758003

1 Introduction

In the field of Human-Computer Interaction (HCI), as well as in multiple areas of society and academia, there is a call for a radical shift from a human-centered way of interfacing with other beings to a More-than-human (MtH) perspective: It seeks to decenter human perspectives and recognize non-human entities (e.g., things and non-human lifeforms) as (co-)participants in shaping the world. Scholars

*Also with Escola Universitària ERAM Universitat de Girona.
†all authors contributed equally and can therefore list themselves as 2nd authors

This work is licensed under a Creative Commons Attribution 4.0 International License.

Mindtrek '25, Tampere, Finland

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1512-9/25/10
https://doi.org/10.1145/3757980.3758003

such as Haraway [27, 30], Demeritt [16], Puig de la Bellacasa [13], and Barad [4] highlight the interconnectedness of humans and non-humans: They advocate for ethical, relational, and interdependent engagements that recognize our interconnected agencies with non-humans shaping our realities.

The design community has echoed MtH approaches: with the recent MtH turn, designers increasingly adopt attitudes that decenter humans and place non-humans at the center of design [61]. Yet, applying abstract concepts like decentering in practical design contexts remains difficult [23, 43], often requiring designers to interpret them differently [44]. This has led to diverse interpretations—from assigning roles to non-humans (e.g., as co-designers or users [11]) to purposes such as designing for multispecies cohabitation, attunement, or discovering MtH interconnections [44]. While such plurality enriches the field, there is still limited articulation of the nuanced roles and purposes enacted in MtH design. Addressing this is vital not only for unpacking lingering human-centered assumptions in MtH design (e.g., [45]) but also for understanding the implications of MtH philosophies in generative and design practices.

To explore this gap, we conducted a first-person, bottom-up and exploratory reflexive design studio using a research-through-design (RtD) approach [64]. As a team of design researchers, we investigated the deep interconnectedness of humans and fungi [49, 60] through exploring the concept of a mushroom basket metaphorically as an interface between humans and non-humans in a design experiment to explore entanglements between humans and non-humans. We drew on our lived experiences, tacit knowledge, and collective sense-making to guide this process. Through a series of reflexive sessions, we collaboratively unpacked the roles and purposes embedded in our designs. As a result, we contribute a framework that concretizes potential roles (e.g., humans as materials, non-humans as co-makers) and purposes (e.g., activism, collective survival, approximation) in MtH design. While grounded in our Reflexive Design Studio experiment and not intended as a definitive account, this framework emphasizes the exploratory nature of our contribution. It highlights how MtH complexities can be surfaced through conceptual design work, offering lenses that others can take up, test in situated practices, contest through alternative interpretations, and expand into new directions. In doing so, we provide critical and generative starting points for future MtH design research that seeks to bridge ambitious theoretical aspirations of MtH approaches with the nuanced realities of design practice.

2 Background: More-Than-Human (MtH)

MtH approaches draw heavily from post-humanist and feminist philosophies that seek to decenter anthropocentric perspectives, calling for a recognition of non-human entities as co-participants in shaping the world. For example, Haraway's concepts of the cyborg [27] and companion species [30] challenge human exceptionalism by emphasizing mutual interdependence between humans, machines, and non-human life. Demeritt [16] similarly critiques binary divisions like human/nature, framing humans and non-humans as co-constituted within socially constructed systems. Barad's notion of intra-action [4] further destabilizes such dichotomies, proposing that beings emerge through dynamic material engagements rather than pre-existing identities. Building on this, Puig de la Bellacasa [13] introduces "matters of care" as an ethical and political practice that binds humans, non-humans, and environments in transformative relations that challenge dominant power structures. Finally, Tsing's work on matsutake mushrooms [60] illustrates multispecies interdependence amid ecological damage. Her notion of "noticing" underscores how attentiveness to human–non-human entanglements challenges dominant narratives of human control. Together, these scholars provide a theoretical foundation for MtH design by urging us to move away from human-centered frameworks toward a more relational, ethical, and interdependent approach, and to actively consider ourselves as existing in MtH worlds.

3 Related Work: MtH Design

The philosophical shift of MtH is reflected in design, where MtH design frameworks challenge traditional human-centered approaches, acknowledging non-human actors —— such as things, animals, plants, and microorganisms— as active participants in the design and post-design processes [61]. In a broader overview, for instance, Coskun et al. [11] outline MtH design as designing for and with non-humans, framing non-human actors as users or (co-)designers. Exemplifying the co-design aspect, Dolejšová et al. [18] explore the properties and agencies of food ingredients and microbes in human-food interaction design. Behzad et al. [5] approach MtH design through a crafting-based practice, framing the ways ceramics break and decay as non-human participation design. Genç et al. [24] similarly treat fungal behavior as a co-creative force in shaping the outcomes of bio-interfaces. Extending these situated approaches, Wakkary et al. [63] propose 'backyard practices' as liminal engagements with more-than-human worlds, foregrounding everyday sites as spaces for acknowledging relations between humans, plants, animals, and materials as part-takers in design. Furthermore, in animal-computer interaction (ACI), non-human animals are approached as users whose preferences are elicited through material engagement —e.g., through toy interaction [3, 9] or inter-animal communication systems [33]. These approaches all present iterative design processes, based on understanding the needs and influences of non-human others in design through multiple trials, errors, and careful observations in each iteration.

Many MtH design works extend beyond assigning roles and use design to explore broader purposes such as multispecies cohabitation, attunement, or care. Kneile et al. [34], for example, review post-anthropocentric artifacts and identify design strategies across three categories: exploring and uncovering non-humans; living together and understanding each other; and empowering non-human entities. These strategies are proposed to pursue goals ranging from addressing human-made environmental concerns to exploring curiosities about encounters with non-human nature. Similarly, Nicenboim et al. Nicenboim et al. [44] highlight that "decentering in design" can involve diverse goals—including decentering human privilege in favor of multispecies cohabitation, or uncovering and making visible interconnections and problematic power relations. Exemplifying these, Sondergaard et al. [54] and Bell et al. [7] engage biological materials and feminist theory to reconfigure dominant human-centered perceptions of bodies through practices like feeding soil with menstrual blood, crafting pads from soil-grown

biomaterials, or growing interfaces from human skin microbes to highlight the body as a MtH biome. Tomico et al. [58] explore human–plant cohabitation through spaces that entwine daily life, and later [56] extend this by conceptualizing 'constituency' via relocating a plant design studio, showing how decentering emerges by shifting who and what matters in design.

In MtH design, working with biomaterials has become a key site for exploring human–non-human relations, and mediation of them through design practices. Ikeya et al. [31] review critical perspectives in this area and identify three tendencies: mediating human understanding of biomaterials, challenging ethico-political biases, and engaging with biomaterials relationally across settings from the lab to the wild. Liu et al. Liu et al. [38] and Rosén et al. Rosén et al. [50] focus on approximation and mutual survival, introducing tools like soil sensors and wearable sticks to help humans sense and support non-human flourishing. Similarly, Bell et al. [6] design calendars that visualize microbial temporalities, reconfiguring human routines through more-than-human temporal frameworks. Similarly, Oktay et al. [46] emphasize the temporality of care in more-than-human design, showing how attention to unfolding rhythms and durations of different non-humans can reshape design engagements. Complementing these emphases on care and collective survival, Altarriba Bertran et al. [1] propose a taxonomy of interactive technologies for joyful human–forest interactions, framing play and joy as key modalities for more-than-human attunement. Resonating with these works, Tomico et al. [57] offer a repository of MtH design methods that foreground sensitizing humans to MtH contexts and fostering human–non-human relations through material engagement. These examples show how MtH design not only reassigns roles but interrogates ethical, ecological, and existential purposes.

While these works highlight the diverse ways MtH stances are applied in design, they also surface the challenge of operationalizing abstract MtH premises in design practices [23, 43]: Designers actively interpret and adapt them differently within the complexities of their situated contexts. The diversity in interpreting MtH philosophies, as we see it, stems from the inherently situated nature of design: Drawing on Haraway's concept of situated knowledges [28], we argue that design is always shaped by specific positionalities, and mediated by designers' experiences, cultural backgrounds, and societal influences. The intricate interplay of theoretical aspirations, such as decentering humans and embracing relationality, with designers' positionalities and situated contexts, reveals an ongoing negotiation between abstract ideals and the realities of design practices: E.g., even when decentering human perspectives, MtH design engagements sometimes surfaces human-biases, as shown by works like Ofer and Alistar's Ofer and Alistar [45] where efforts to care for microorganisms still reflected human desires for creative expression.

Here, it becomes evident that, while MtH design draws upon rich philosophical foundations that challenge anthropocentric perspectives, the practical application of these ideas in design remains complex and highly situated. We believe that articulating a comprehensive list of the different nuances in interpreting MtH thought in design practices can contribute valuable starting points to generatively and critically examine how theoretical MtH principles are interpreted and enacted in practice. This approach may serve as a bridge between the philosophical imperatives of MtH thinking and their situated manifestations within the complexities of design. Our work directly addresses this gap by concretizing roles and purposes as pivotal to the negotiation between abstract MtH ideals and the situated realities of design practice.

4 Method: Reflexive Design Studio

This work seeks to surface nuanced ways MtH thoughts may surface in design practices: we employed an approach, which we call a *Reflexive Design Studio* — a RtD-informed [64], hands-on method for producing knowledge through design practice. The Design Studio is a long-standing approach rooted in both design pedagogy [53] and industrial practice [55], characterized by open-ended briefs and a culture of iterative critique and feedback [8, 35]. In our experiment, we embraced this tradition by collectively defining a brief, sketching conceptual designs individually, and engaging in ongoing collective discussions to iteratively reflect on and refine our design approaches. Unlike traditional studios that emphasize innovation or problem-solving [55], our studio centered on surfacing and challenging the roles and purposes we enacted in design. Through a process of deconstruction [17], we examined our biases and intentions by confronting contradictions, reframing assumptions, and generating alternative concepts that foreground MtH entanglements.

To explore these in an actionable way, we use the pluralistic relationship between people and mushrooms as a case: from beer making to impacts on human health, psychedelics, and even unwanted bathroom growths, humans and mushrooms are deeply intertwined [49]. Their (un)wanted ubiquity makes them compelling companions, actors, and antagonists [21]. Human–mushroom interconnection has long been explored in MtH research, where mushrooms are imagined as participants—even protagonists—to reveal multispecies interdependencies [48, 59, 60]. Moreover, the interconnected, non-hierarchical ways mushrooms live offer a blueprint for examining MtH relationalities. Several studies have adopted mushrooms as a framework to conceptualize this interconnectedness [14, 21]. In our case, investigating human–mushroom relationalities through design provided a fruitful approach for examining such interconnections, and identifying how roles and purposes in MtH design might concretize biases and assumptions in design practices.

4.1 Epistemology

In this paper, we present an account of situated knowledge constructed through the entanglement perspective [22]. This entails that the knowledge presented in this work is shaped by the authors' own lived experiences, and our assumptions are embedded within personal, cultural, and societal discourses: We claim that our hands-on design research produces "situated knowledge(s)", as framed by Haraway [28], the science we propose is a fundamentally human endeavor, characterized by negotiation and plurality. Embracing this "relational" epistemological perspective [25], we adopt an "entangled" worldview [22]: We recognize the intricate connections among humans, non-humans, technologies, and environments. As designers working within these relational ontologies, we see it as our responsibility not only to acknowledge these dependencies, but to engage with(in) them—actively and accountably.

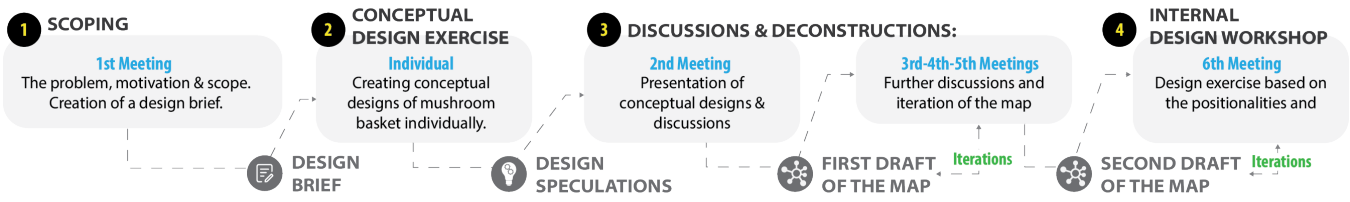

Figure 2: An overview of our Reflexive Design Studio process.

4.2 Statement of Positionality

Our own lived experiences were vital for the work undertaken in this paper. Here, we wish to explicitly acknowledge the influence of our positionalities and tacit knowledges [51]: We operationalise the term, tacit knowledges, to describe skill acquisition and mastery through non-formally recorded processes, e.g., our (design) sensibilities, practices and approaches: Our team comprises six researchers, all in their thirties, five working at Tampere University (Finland) and one working at University of Tartu (Estonia). Five of us have design backgrounds —having both studied and worked in industry—; one has a background in economics. We are an international group from five countries and cultures of origin (Turkey, Spain, South Korea, Germany, and Finland), identifying as four men, one non-binary person, and one person who prefers not to state their gender identity.

MtH, as a concept, plays a visible role in our everyday research and conversations: All of our research revolves around gameful and playful engagements in MtH worlds. While we share an understanding of humankind as part of nature, our perspectives diverge: For example, Velvet draws from feminist HCI and video game analysis; Caglar explores human-biomaterial relations; Ferran investigates situated forest-based design methods; Oz focuses on posthuman play and augmentation. Across our differences, we support MtH design's potential but are cautious of superficial engagements or exploitative misuses of its ideals. We are also, quite practically, a group of mushroom enthusiasts—often foraging, cooking, and referencing mushrooms in our design conversations. As such, we consider ourselves a group of mushroom aficionados, both on a personal and professional level: Throughout our discussions for this paper, and beyond, we regularly use the mushroom (and its associated concepts) as metaphors for inspiring, explaining, and making sense of our research. Examples of influences include psychedelic experiences with psilocybin *(see [39])*, visualizing interlinked complexities as rhizomes (after Guattari and Deleuze [15]), or understanding mushrooms as witnesses and living archives of human-made catastrophes (e.g., nuclear fallout [60]).

4.3 Procedure

We started our process with a meeting to define the study's scope and establish a shared understanding (*Phase 1*, see also Fig. 2). We exchanged our ideas on MtH interplay in design and discussed questions, such as *"How do we define the boundaries between human and non-human entities? Who decides what is designed? Who benefits?"* as loose guides for our process. We agreed to use the mushroom basket as both a case study and a metaphor for exploring MtH relationalities. Rather than strictly referencing foraging, the basket acted as a metaphorical interface between humans and mushrooms. This ambiguity was intentional, echoing the Fuzzy Front-End in design, where opportunities and concepts emerge amid uncertainty and limited context. Upon the first meeting, we individually speculated on mushroom basket ideas for a month (*Phase 2*). Each of us was free to represent concepts via sketches and/or verbal descriptions. Each author presented at least three ideas, which in total summed up to 30 more-or-less refined conceptual designs.

In *Phase 3*, we came together to present and critique each other's concepts using a Miro board. Drawing on Derrida's notion of deconstruction [17], we unpacked ingrained assumptions —such as the prioritization of humans— and reflected on how these shaped MtH entanglements in design. Upon this, the first author drafted a map by clustering themes from the discussion (similar to affinity diagramming [40]). The map reflected our concepts and discussions: For example, by roles like "humans as materials" and "non-humans as co-makers" (5.1 Roles in Design) were based on the designs and discussions on challenging hierarchies in MtH design, where as purposes such as "activism" and "collective survival" (5.2.1 Survival) were created based on concepts and discussions on uncovering tensions between human agency and non-human flourishing. We refined the map over three follow-up meetings, generating new ideas, roles, and purposes. This process culminated in a shared text and a set of design cards with short descriptions of each role and purpose, along with examples that helped us formulate them.

After our fifth meeting (*Phase 4)*, we held a two-hour internal design workshop in two pairs. Each group selected one role and one purpose from different categories of our MtH map (see Findings) as starting points. These were used as reflective probes rather than fixed constraints—sometimes intentionally subverted (e.g., imagining humans as material for mushrooms). The workshop yielded six new concepts and surfaced an additional role. We reflected on these concepts and their implications, refining our role and purpose descriptions and adding comparable MtH design work as examples.

The final map is created through multiple iterations of this process, including peer-review-driven revisions. As part of this, we removed human-centered roles or purposes —such as "non-humans as materials" or "MtH as solo-surviving for humans"- we felt contradicted MtH values.

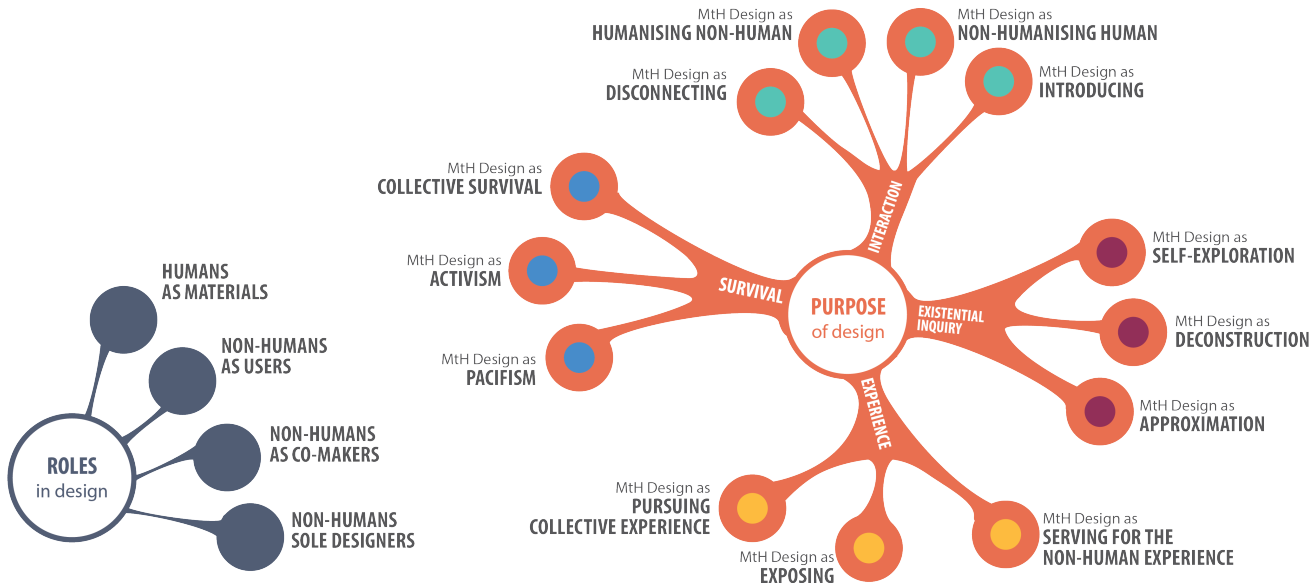

Figure 3: Map of Roles and Purposes in MtH Design formulated based on our Reflexive Design Studio.

5 Findings: A Map of Roles and Purposes in MtH Design

Through our method, we developed a map of roles and purposes that captures how MtH thinking was operationalised in our Reflexive Design Studio. The map (Fig. 3) consists of two main categories: (1) Roles in MtH Design and (2) Purpose of MtH Design. We now detail this map, grounded in our design discussions and reflections.

5.1 Roles in Design

The roles category elaborates on how humans frame non-human and human participation in the design activity. The roles presented under this category range from considering non-humans as co-makers or users to extreme stances where the human embraces the non-human agency to a point where humans exclude themselves from the process:

"Humans as Materials" positions human bodies and minds as sites of material entanglement with non-humans. The concept of "hallucinogenic mushrooms" (Fig. 4a), a wearable with mushroom growing over human skin by using its nutrients (i.e., sweat and microbes), illustrates an example design outcome for this role. This is similar to Sondergaard et al.'s menstrual product designs [54] that, e.g., utilise humans' menstrual blood as a nutritious material for soil. Aligning with these, this role attempts to transform the dominant human-centered power relations by reframing humans as part of MtH existence, contributing to the non-human well-being by offering themselves as materials for non-humans.

"Non-humans as Users" emphasises understanding the needs of the mushrooms and helping them nurture through designs. We created this role based on the *"city shroom bot"* (Fig. 4b). This concept describes a robot on which the mushrooms grow. The sensors on it detect the habitable environments in the city and move toward those places where mushrooms can grow easily, similar to biohybrid robots [12] that combine robotics with mushrooms, adorning them with mobility features. This role can focus on cultivating relationalities by prioritizing the non-human perspectives, positioning humans as facilitators rather than dominators. Through iterative material engagements with non-humans (e.g., by creating design artifacts and attuning non-humans), humans try to understand and support non-humans' needs, paying attention to how non-humans intra-act with design interventions. From the perspective of MtH design, however, it brings in the assumption that "we know what non-humans need," which requires a humble stance, an awareness that we estimate, rather than knowing the needs of the non-humans. Designing interventions to accompany the needs of non-humans calls for a responsible trial and error, as in Brown et al.'s case of designing animal feeders [9], as well as an effort to attune and understand the ways non-humans exist (e.g., their temporalities [6]) and a constant reflection and contextualization of the designers' decisions in relation to other practices of humans and non-humans [62].

"Non-humans as Co-makers" frames mushrooms as active actors in the design process together with humans. *"Fashionable baskets"* (Fig. 4c) and the *"mycelium expression basket"* (Fig. 4d) were the concepts that helped us frame this role: The *"fashionable basket"* imagines a set of half-designed garments, the final form of which is defined by the mushroom growth. Similarly, the latter concept proposes a half-designed mushroom-habitable retroreflective cloth, a cloth that reflects the light strongly. The design process continues when it is placed in nature, where mushrooms cover the surface of the fabric by growing over it. Human-directed light to find it might reflect back, or it might not, depending on the surface coverage due to the growth of the mushrooms. In both, the MtH relationality is built through shared agencies: human intentions are intertwined with non-human growth patterns, and the final designs

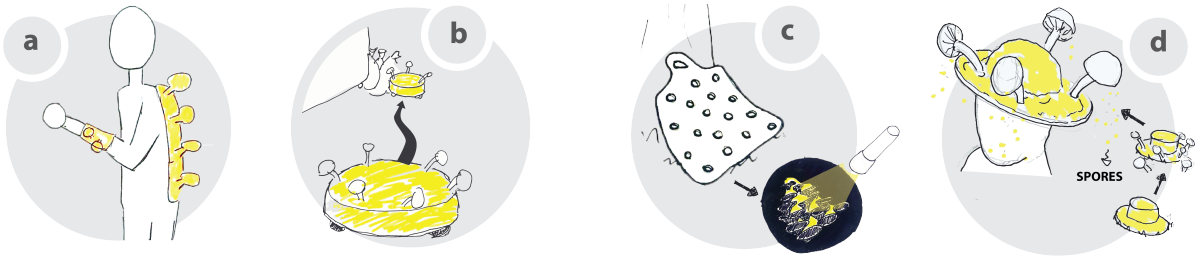

Figure 4: Illustrations of design concepts: (a) "Hallucinogenic Mushrooms", a wearable, growing mushrooms using human skin microbes; (b) "City Shroom Bot", a robot that moves mushrooms to habitable places; (c) "Fashionable Baskets", a hat shaped by fungal growth; (d) "Mycelium Expression", a habitable fabric that reflects light if mushroom does not cover it.

are shaped neither solely by humans nor by mushrooms but by their entangled actions. This approach is similar to Behzad et al.'s exploration of repertoires for design iteration in MtH design [5], in which the authors considered the breakage of ceramic material as the active influence in defining the designs. As exemplified in these concepts, this role pursues a curious exploration about how we, as humans, can actively relate to non-humans through discovering our entangled influences in design and giving more space for non-humans to lead the designs.

"Non-humans as Sole Designers" is rather an extreme end of leaving a space for non-human agency in design processes to a point where humans reduce their role to observers. This role was created based on our discussions about considering growth as the sole agency of mushrooms in the making processes. We asked questions such as "What would mushroom design with no human involvement at all?" This led us to consider the structures mushrooms create in nature (i.e., forms, textures, and colors of the ground mushrooms or mycelium webs underground) as the design activity of mushrooms without any physical intervention from humans. As such, there is still an emergent agency between humans and mushrooms as humans observe the ways mushrooms grow, but humans, rather than physically intervening, situate themselves within non-human contexts to formulate design insights. This is similar to the concept of biomimicry in design [32], drawing design inspiration from nature's forms, processes, and systems, yet in MtH design contexts, considering non-humans as sole designers differs from this concept in a way that it does not necessarily aim to utilise the design knowledge for creating more designs.

5.2 Purposes of Design

The purpose of design refers to its intended end goal. We identified four themes: (1) species survival, (2) human and non-human experiences, (3) interactions among humans, mushrooms, and other non-humans, and (4) existential inquiry into ontological perspectives. Below, we present the subcategories and specific purposes within each theme, noting that "MtH Design as" is omitted in the purpose titles to avoid repetition: e.g., Activism, instead of MtH Design as Activism.

5.2.1 ***Survival.*** The survival subcategory considers how MtH design might shape practice around anticipated benefits for the survival of humans, mushrooms, or other non-humans. This reflects a pragmatic view of MtH interplay through three purposes identified, addressing different nuances about the flourishing across species:

"Collective Survival" is about embracing a perspective towards supporting the survival of both humans and mushrooms, either by catering to the needs of both directly or by helping humans to be aware of the importance of caring for non-humans alongside humans. An example of the former is the *"symbiotic basket"* (Fig. 5b): a basket that uses computer vision and other advanced sensors to navigate humans to mushrooms, while also sending humans on detours so the spores in the foraging basket are disseminated. The latter approach was illustrated via the *"shaky compass basket"* (Fig. 5c): a basket that guides humans around the forest, like a compass, but only does so if the human keeps shaking it recurrently to distribute spores. In both cases, the relationalities extend beyond foraging to include the ecological implications of supporting mushroom growth. Interactions—like shaking the basket or using sensors—tangibly represent mushroom–human interdependence, evoking care for non-humans while meeting human needs. HCI is increasingly adopting similar perspectives: For example, Tomico et al. [58] reflect on cohabiting with plants—reshaping living spaces, caring for them directly and through technology—to foster collective survival. Liu et al. [38] exemplify design interventions such as interactive vests and sticks that help humans attune, engage, and expand mushroom existence while foraging. Like these examples, MtH design as collective survival envisions tools and systems that foster practices of interdependence, negotiating the needs of both humans and non-humans.

"Activism" frames design as advocating for mushroom survival, even at the expense of humans. One example is *"a fool with a basket"* (Fig. 5d): a foraging basket that misleads humans to walk longer paths, dispersing more spores. Pushing this further, *"humans as yeast baskets"* (Fig. 5e) proposes a skin patch for yeast infections that, instead of healing, spreads scents to attract touch and propagate yeast across bodies. These concepts illustrate activist stances where non-human thriving overrides human benefit. Similar approaches appear in sustainability design, such as Marttila's [41] "unpleasurable" products and interfaces that provoke discomfort to expose human-caused harms. In this light, activism in MtH design positions relationality as advocacy, using material encounters to provoke reflection or unease for the sake of non-human survival.

"Pacifism" embraces a "live and let others live" ethos, where designers step back to avoid imposing on non-humans and let them

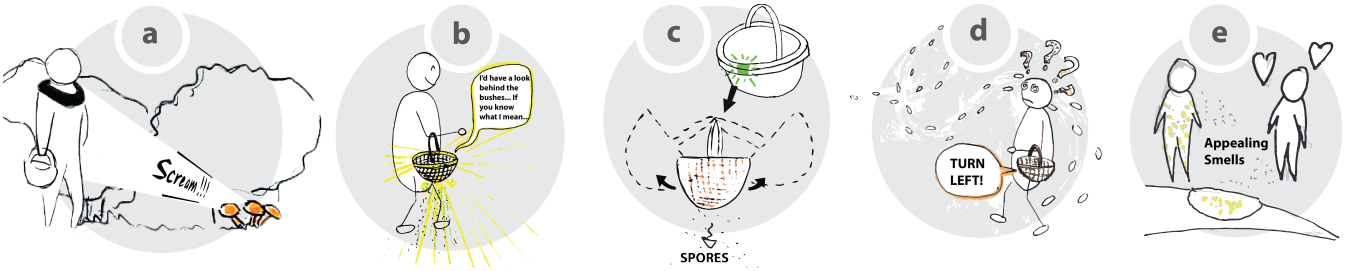

Figure 5: Illustrations of design concepts: (a) "Screaming Mushrooms", a wearable that emits 'scream-like' sounds from the direction of mushrooms; (b) "Symbiotic Nasket" gives ambiguous directions; (c) "Shaky Compass Basket" shows the direction of mushrooms when shaken and spores are distributed; (d) "a Fool with a Basket" misleads human by giving incorrect directions; (e) "Humans as Yeast Baskets", a skin patch that distributes smells to attract others into touching and promoting yeast propagation.

exist on their own terms. This stance emerged from our discussions about not designing at all if it risks harming non-humans, framing pacifism as respect for their autonomy. Taking this stance is not inactivity but a deliberate design choice. Echoing Sandelin and Homewood's "Design (In)actions" [52], pacifism means making informed decisions not to design, as exemplified by the "Place Léon Aucoc" project in Bordeaux, where architects concluded no redesign was needed beyond small interventions like more frequent cleaning. Similarly, MtH pacifism involves observing and engaging with non-humans while recognizing the value of non-intervention to avoid disrupting their processes.

5.2.2 ***Experience.*** Unlike the practical focus of survival, this subcategory centers on experiences. Experience-oriented design highlights experiential dimensions such as hedonic and eudaimonic aspects [42]. In MtH design, we explored purposes aimed at momentary joys (hedonic) or longer-term goals (eudaimonic), such as ethical happiness over time. Because our interventions mainly involved mushrooms, we often assumed we could estimate non-human experiences—raising questions like: Do they have virtues such as competence or desires, or are these human projections? And even if such virtues exist, how can their experiences be evaluated? The purposes here reflect these questions.

"Pursuing Collective Experiences" aims to facilitate shared, co-constituted experiences between humans and non-humans. In our mushroom context, we struggled to create such concepts, as interactions with mushrooms—being relatively alien to humans—are harder to frame beyond growth or spore dispersal. However, this goal is more evident in interactive systems for playful pet–human relations. For example, Pons et al. [47] describe mixed-reality games where humans trigger toys for hamsters or play tablet games with cats catching sea creatures. Such designs illustrate how MtH approaches might focus on co-constitutive participation, where human and non-human boundaries blur through non-utilitarian encounters like play. Here, we still noted that it is crucial not to assume that non-humans necessarily enjoy these interactions; their behaviors may arise from instinct rather than enjoyment.

"Serving Non-human Experiences" extends the previous purpose by centering design on creating "favorable" experiences for non-humans. While we found no mushroom-specific examples, parallels exist in Animal-Computer Interaction, where designs prioritize playful animal experiences. For example, Andreou et al. [3] created enrichment toys for bottlenose dolphins, informed by caregiver input and dolphin interactions. Such work reorients design around non-human perspectives, casting humans as curious facilitators rather than beneficiaries. Material encounters thus become sites where humans try out their intentions to provide joy to non-humans and observe how they turn out. While this reflects a move away from anthropocentrism toward prioritizing non-human experience —such as replicating playful hunting for dolphins— evaluation still relies on human observation. Ultimately, claims about creating "joyful" or "playful" non-human experiences remain speculative.

"Exposing" aims to lessen human agency by foregrounding non-human influence. We created this purpose based on the concepts of *"hallucinogenic mushrooms"* (Fig. 4a) and *"humans as yeast baskets"* (Fig. 5e), as well as *"inoculation wearable"* (Fig. 6b). The latter is a wearable that induces yeasts on the skin and can put human users in a passive state where the experience is driven by the mushrooms' agency whereas the prior examples do the same by giving mushrooms the autonomy to induce hallucinogenic experiences in humans or to seduce other humans into experiencing yeast infection. Overall, this purpose enacts relationalities that diminish human dominance, allowing non-human agencies to take the lead in shaping experiences. Together, these designs enact relationalities that diminish human dominance, positioning bodies as sites of non-human influence. By creating experiences that may be unpleasant or disorienting, "exposing" destabilizes human control and prompts reflection on entangled human–non-human relations.

5.2.3 ***Interaction.*** In the interaction subcategory, we introduce purposes that regulate the channels of communication between mushrooms and humans. We see this communication as already present —e.g., through growing, foraging, observing, or eating them. However, our ideas from the concept generation phase revealed more elaborate concepts where the design can execute the design activity for connecting humans and mushrooms in various ways, e.g., through different (human or non-human) modalities. Further discussions on the subject also let us understand some design perspectives on diminishing or openning communication channels among non-humans and humans. Here we present four purposes that MtH design practice can embrace:

Figure 6: Illustrations of design concepts: (a) "Feeling Mushrooms" translates electrical pulses of mycelium for humans; (b) "Inoculation Wearable" induces yeast to human; (c) "Camouflaging Mushrooms", shoes that project visuals to hide mushrooms from humans; (d) "Worming Mushrooms", shoes that leave organic traces, guiding worms to mushrooms.

"Humanising Non-human" frames design through anthropomorphism, as in *"Screaming Mushrooms"* (Fig. 5a), where mushrooms scream to guide foragers. Here, mushroom signals adopt human modes of communication to become more legible. Anthropomorphism is common in human–robot interaction (e.g., giving robots human traits [20]) and also appears in human–animal interaction design, such as AI narrators that translate cat videos into verbal messages [36]. While such approaches can foster empathy toward non-humans, they risk oversimplifying or misrepresenting non-human agencies by privileging human perception. This tension—making non-humans relatable while erasing their distinct ways of being—raises critical ethical concerns about anthropocentric exclusions in design [19, 26, 37].

"Non-humanising Human" takes the opposite stance, inviting humans to adopt non-human modalities. Reflections on our *"Feeling Mushrooms"* concept (Fig. 6a)—an artificial finger skin that translates mycelium signals into small human-perceivable currents—illustrate this approach. Rather than making mushrooms more human-like, the design extends human perception to attune to non-human pulses on the go. Similar strategies appear in MtH design, such as soil sensors that let humans feel soil states [38, 50] or interfaces translating plants' [26] or bacteria colonies' [6] states into sound. These works show how such designs expand human perception toward non-human temporalities and modalities. Yet, their reliance on technological mediation risks oversimplifying non-human complexities and reducing possibilities for richer, direct intra-actions (e.g., touching soil [50]).

"Introducing" shifts the focus from human–non-human to non-human–non-human interactions. This stance emerged in our discussions of humans creating bridges between habitats, enabling species encounters. We illustrated it with the *"worming mushrooms"* (Fig. 6d): shoes that leave organic traces to guide worms toward mushrooms. Similar efforts appear in HCI, such as Kleinberger et al.'s [33] parrot-to-parrot video-calling system. Embodying this purpose, MtH design intervenes to facilitate encounters and communication among non-humans.

"Disconnecting" seeks to block encounters between humans and mushrooms. Contrary to MtH approaches that emphasize fostering relationalities (e.g., [54]), this purpose reflects a pessimism toward human impact, suggesting mushrooms thrive better out of human reach. We illustrated it with *"camouflaging mushrooms"* (Fig. 6c): shoes projecting visuals to hide mushrooms in urban spaces. The relation here is one of protective distance—humans are excluded to safeguard non-human flourishing. By actively limiting engagement, this purpose reframes agency as resistance, acknowledging that not all MtH encounters are desirable.

5.2.4 ***Existential Inquiry.*** In this subcategory, we outline ontological perspectives emerging from our design experiments. Here, the designer re-frames their own body or mind as part of MtH engagements with mushrooms—potentially becoming the mushroom basket themselves. These purposes center on philosophical and fundamental concerns.

"Self-exploration" frames the mushroom basket as a tool for self-discovery, where the body and mind are seen as vessels or surfaces for mushrooms—from carrying them in the stomach or hands to yeast infections. This framing extends beyond design artifacts to processes and actors, like doctors or mirrors, that help unpack human–mushroom entanglements. In our context, this was illustrated by *"carrying mushrooms"* (Fig. 7e), a full-body suit that higlights fungi-dense parts of the body through vibration and visualisation. A parallel can be seen in Bell et al.'s work with human microbiomes [7], where materials grown from microbiome samples revealed how daily activities (e.g., exercising, hugging, typing) affect microbial life. Such approaches highlight how MtH design can probe existing body–mind entanglements with MtH worlds.

"Deconstruction" focuses on recognizing humans and mushrooms as interconnected lifeforms within ecosystems. Unlike self-exploration, which centers on the individual, this approach shifts scale to make entanglements visible at an ecosystem level. MtH design here questions the boundaries between humans and mushrooms, reflecting on their similarities and differences through (often surreal) acts of deconstruction. In our discussions, we illustrated this with an art installation portraying mushrooms as holes in a forest (Fig. 7b), where their removal invites reflection on human–mushroom interdependence. In a similar vein, but reversing the strategy from reflecting on the absence of non-humans to unpacking interconnections of humans and non-humans, Sondergaard et al. present their exploration of biomaterial-based menstrual artifacts [54]: Throughout a series of participatory crafting sessions and experiments with biomaterials, they discovered how human menstrual cycles could be reframed from stigmatising these processes to considering menstrual cycles as tools that we discover the blurred boundaries of human bodies to broader MtH worlds, e.g., by designing menstrual pads that nurture soil via menstrual blood [54]. Together, these perspectives exemplify deconstruction

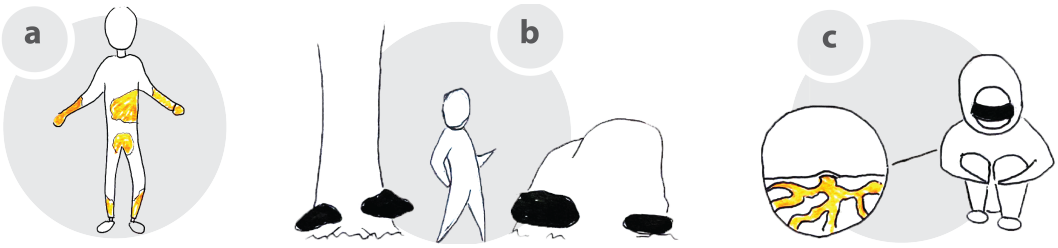

Figure 7: Illustrations of design concepts: (a) "Carrying Mushrooms", a suit that visualizes and vibrates on fungi-dense areas; (b) "Absence of Mushrooms", an art installation that replaces mushrooms with black bags to highlight their absence; (c) "Becoming Mushroom", a suit simulating somatic sensations for humans to experience becoming a mushroom fruiting body.

as a design stance that foregrounds our entanglement with broader MtH ecologies.

"Approximation" seeks to connect with mushrooms by adopting their ways of being, such as perceptions of time, space, and activity. This perspective emerged from our discussions on human versus mushroom temporalities and was illustrated by the *"becoming mushroom"* concept (Fig. 7c): an extended-reality suit simulating the emergence of a fruiting body through sensations like soil pressure, warmth of light, and restricted movement. Such designs encourage role-play, mimicry, or simulation to experience non-human perspectives. Aligned with this, Cosio et al. [10] created a VR experience where participants embody a flying squirrel, engaging with its behaviors and environment. These examples show how approximation bridges human and non-human ways of being through embodied simulation.

6 Discussion

In this paper, we explored the MtH design through a Reflexive Design Studio, surfacing roles and purposes embedded in our practice. Unlike prior work, our study uncovers a map of different practical MtH design nuances, not as fixed definitions of MtH interplay, but as starting points to generatively and critically explore MtH in design practices. We argue that the framework presented here offers relevant —and at times contradictory— yet necessarily inconclusive stances on how MtH can be interpreted in design practice. The roles and purposes we outline are not detached from prior work but wrestle with existing scholarship, practices, and approaches. Alongside efforts to conceptualize MtH design (e.g., the broader framings by Coskun et al. [11] and Nicenboim et al. [44]), or the many similar situated MtH design practices (examplified in the Findings sections while presenting roles and purpose), our framework makes visible the frictions and tensions in MtH practice by concretizing them —provocatively— as roles and purposes that "stay with the trouble" [29]: we elaborated and at times exaggerated the specifics and nuances revealed through our Reflexive Design Studio. By framing these as one set among many possible interpretations, we invite others to extend, question, or reconfigure them toward more robust, transparent, and collective applied MtH design practices.

In the following sections, we first discuss how our possible roles and purposes resonate, conflict, and/or expand the understanding of MtH design by surfacing practical nuances in design, then articulate how our framework and methodological insights might be taken up by future MtH design research work:

6.1 Collective Becomings through MtH Design

Karen Barad's concept of becoming [4] offers a relational ontology in which entities do not preexist interactions but emerge through ongoing intra-actions. This challenges static, human-centered design by emphasizing the dynamic, co-constitutive processes connecting humans and non-humans, framing MtH design practices and outcomes as sites of entanglement and mutual constitution. Our roles of *"Non-Humans as Co-Makers"* and the purpose of *"Collective Survival"* align closely with Barad's concept of becoming: *"Non-Humans as Co-Makers"* highlights the shared agency in design processes, where non-human entities actively shape outcomes. This resonates strongly with works such as Wakkary et al.'s "thing-centered design" [61], Tomico et al.'s constituency as a Matter of Practice [56], and Coskun et al.'s "designing with" non-humans [11], emphasising shared agency and mutual influence in design. Likewise, *"Collective Survival"* focuses on care-based intra-actions, where humans and non-humans co-constitute each other's survival through interdependence. Tomico et al.'s reflections on living with plants [58] and Liu et al.'s collaborative survival [38] similarly highlight how survival can be understood as co-constituted within and through designs.

"Pursuing Collective Experiences", however, introduces a novel dimension by foregrounding non-utilitarian, experiential encounter —particularly play— between humans and non-humans as a way of becoming. While existing practice-based works rarely address this aspect (with exceptions such as Bertran et al.'s joyful forest technologies [1, 2]), our framework explicitly positions play and other open-ended encounters as sites of collective becoming. In this sense, our work extends prior MtH design work by suggesting that MtH design can aim collective becomings not only by focusing on collective survival and cohabitation but also with forms of joy, curiosity, and companionship. It aligns with Haraway's companion species [30] where she highlights instances of humans and dogs materially becoming together in the acts of play. In this regard, *Pursuing Collective Experiences* brings in possibilities of MtH design approaches to actively engage and design for becomings through experiences such as play.

Together, these roles and purposes show how MtH design can enact becoming not only through the act of designing but also through the designs themselves. They foreground the potential of MtH design to materialize entangled relationships and facilitate collective becoming, while also exposing a challenge: designing

for *"Collective Experiences"* requires avoiding anthropocentric assumptions—such as presuming non-humans enjoy experiences as humans do. Instead, designers might prioritize open-ended designs that sustain complexity and allow human–non-human agencies and relationalities to emerge on their own terms.

6.2 Non-human Flourishing as an Ethical Commitment in MtH design

MtH approaches highlight the need to challenge anthropocentric narratives of control and progress, stressing human responsibility toward non-human flourishing in the pursuit of multispecies thriving (e.g., see De la Bellacasa [13] and Tsing [60]). Our roles and purposes offer ways to operationalize this in design. For example, *"Non-Humans as Users"* and *"Serving Non-Human Experiences"* underscore the value of situated, iterative engagements in uncovering and addressing non-human needs while challenging anthropocentric assumptions about design outcomes serving human needs. These also resonate with Coskun et al.'s call to design for non-humans [11] and Brown et al.'s view of iterative, trial-and-error approaches attuned to non-human temporalities [9]. However, our framework also extends these perspectives by introducing more radical roles and purposes that push ethical commitments beyond conventional framings of care: For example, *"Humans as Materials"* repositions the human body-mind as a resource to serve non-humans, while *"Activism"* provokes discomfort to humans for confronting anthropocentrism, and *"Disconnecting"* withholds encounters altogether to prevent human harm. Together, these highlight the practical nuances that position care toward advocacy and disruption, expanding the ethical terrain of MtH design in practice.

On the other hand, roles like *"Non-Humans as Users"* may risk projecting assumptions about non-human needs. As de la Bellacasa [13] reminds us, care requires reflexivity and awareness of the limits of human understanding. Furthermore, as Oktay et al. [46] also caution, care is not only about serving needs but about sustaining relationships across overlapping temporalities, where caregiver and care-receiver roles remain fluid. Here, designers must remain vigilant against projecting human-centric biases onto non-human needs and responses. Moreover, the *"Activism"* stance of prioritizing non-human survival over humans and *"Disconnecting"* humans and non-humans to prevent human harm may conflict with Tsing's emphasis on fostering coexistence [59]. In this regard, these purposes may alienate humans while aiming to transform existing anthropocentric power relations. We must carefully navigate this balance between prioritizing non-human flourishing and maintaining human engagement and commitment.

6.3 Mediating Encounters through MtH design

Anna Tsing [60] frames *noticing* as an embodied, relational practice grounded in material engagements between humans and non-humans. Rather than passive observation, noticing is active, situated, and encounter-driven. In MtH design, noticing and material engagements are highly emphasised for attuning designers to interdependencies (e.g., Rosén et al. [50]; Tomico et al. [57]).

Building on these, our framework exemplifies how attunement itself can become a purpose of MtH design—not only a methodological approach—by designing interventions that mediate encounters among humans and non-humans to bridge the gaps in communication, perception, or experiential existence among MtH entities. In that sense, our work resonates with prior work such as Ikeya et al.'s review of biodesigns that translate non-human responses into perceptible forms [31] and Kneile et al.'s scoping review of post-anthropocentric artifacts for mediation of human-nature interactions [34], which identify design strategies about "exploring and uncovering" and "empowering" non-human entities. For instance, our purposes of *"Humanizing Non-Humans"* and *"Non-Humanizing Humans"* parallel these mediation approaches by making non-human signals perceptible. Differently though, the *"Approximation"* provides further nuances, such as designing embodied simulations of non-human perspectives to "explore" non-human perspectives. Similarly, *"Activism"* is aligned with the emphasis of Kneile et al. on "empowering" non-humans to disrupt human lives [34], *"Disconnecting"* comes as a strategy where MtH design explicitly disrupts the MTH encounters to prevent human harm on non-humans. Furthermore, our works articulate more reflexive roles and purposes that place humans engaging with designs in a position to reflect on their own entanglements —such as through *Self-exploration*, which foregrounds intimate relationalities, or *Deconstruction*, which makes broader ecological interdependencies visible. In this way, our roles and purposes provide practical nuances that not only mediate encounters but also become instruments for unsettling, triggering reflections on, and re-situating human positionalities within MtH worlds.

Together, these interpretations resonate with noticing by illustrating design's capacity to mediate relational understanding, while presenting nuanced approaches that balance sensory reorientation, embodied simulations, and reflexive engagements across scales. These roles and purposes, however, also uncover tensions that challenge the practical and ethical boundaries of MtH design: *Humanizing non-Humans*, for instance, may risk oversimplifying non-human agencies and reinforcing anthropocentric biases. Similarly, the reliance on technological mediation in *Non-humanizing humans* may reduce the richness of direct human-non-human encounters. The simulations through *Approximation* may inherently simplify complex non-human experiences, prompting critical reflection on the limits of approximation in fostering genuine encounters. *Deconstruction* and *self-exploration* may risk over-generalizing or instrumentalizing human-non-human relationships. *Deconstruction*'s focus on ecological systems may obscure the specificity of individual entities, while *Self-exploration*'s intimate focus might privilege human-centric narratives. These tensions underscore the challenges of designing for noticing and relational encounters in ways that respect the complexities of the MtH worlds.

6.4 Translating Our Insights into Future Work

In this section, we discuss the insights our framework and methodological approach offer for future MtH design research to take up: We argue that our roles and purposes, although rooted in a specific experiment, hold potential to extend beyond our immediate design context and open exploratory directions for how the

framework might be taken up in other settings of MtH design. For example, educational design studios could adopt the framework to help students navigate the complexities of MtH theories through situated practice—for instance, by experimenting with *"Humans as Materials"* to unsettle anthropocentric assumptions, or by framing *"Approximation"* as a purpose to learn from partial, incomplete understandings of non-human worlds. Sustainability-oriented design teams might draw on purposes such as *"Collective Survival"* to foreground long-term ecological commitments or explore *"Non-humans as Co-makers"* in exploring material innovation processes. Beyond fungi, designers engaging with other living materials such as algae, plants, bacteria, or non-human animals could test how these roles and purposes manifest differently across species—e.g., *"Approximation"* in microbial contexts, or *"Activism"* when designing with threatened species. Similarly, commercial design teams or participatory community workshops could adapt the framework to negotiate multispecies entanglements in sustainable product design or urban cohabitation projects, where *"Collective Survival"* might serve as a critical purpose or *"Pursuing Collective Experiences"* might inspire MtH designs that focus on non-utilitarian encounters—such as joy or play—as a way of fostering new relational modes of cohabitation. In extending the framework into these diverse contexts, new roles and purposes will likely be articulated, reflecting the situated entanglements of different human–non-human worlds. In this way, our framework is not a prescriptive set of guidelines but an exploratory scaffold—one that others can test, contest, and expand in diverse contexts to further navigate the complexities of MtH design practice.

Methodologically, we argue that our findings have potential to extend into other design contexts because of our use of the *mushroom basket as a relational, situated, metaphor* within a *reflexive design practice*: We positioned the basket both as a concrete context for inquiry and as a metaphor for imagining broader human–mushroom relationalities. As a context, it grounded our Reflexive Design Studio by providing a tangible site for generating and reflecting on conceptual designs. As a metaphor, it opened space for speculating on different human-mushroom entanglements, resonating with prior imaginaries that mobilize human–mushroom relations in similar ways (e.g., [60]). Reflexivity was also crucial here, enabling us to iteratively unpack and critique the relationalities surfaced through our inquiry, and to conceptualize how they might extend beyond the immediate context. This dual positioning resonates with Wakkary et al.'s Backyard Practices [63], where the backyard serves both as a lived site of multispecies entanglements and as a metaphor for articulating MtH relationalities in and through design practices. Both approaches show how situated contexts can simultaneously anchor inquiry and act as relational metaphors that travel beyond their settings. Taken together, they point to a methodological direction for MtH design: grounding work in concrete contexts that also function as metaphors, and employing reflexive practices to generate insights that can be reinterpreted across contexts. Future work might extend this approach with other situated metaphors—such as seed banks, compost heaps, or beehives as interfaces of MtH relations—within reflexive design studios to create, critique, and contest different nuances of MtH design.

7 Limitations & Conclusion

Recognizing the situated nature of MtH design, we acknowledge that the roles and purposes we formulated—based on conceptual mushroom basket designs—are not exhaustive and may differ across practices. Designers using other methods, such as hands-on making, long-term engagements with MtH entities, or participatory collaborations, may arrive at different articulations. While our multicultural team aimed to reflect diverse MtH understandings, interpretations may vary further for individuals from different social, cultural, or political backgrounds, or with distinct lived experiences and tacit knowledges (e.g., marginalized, Subaltern, or Indigenous communities [16]).

With these limits in mind, our study contributes a generative and reflexive framework for MtH design by using the mushroom basket metaphor to imagine relationalities beyond human-centered ones, and by employing a Reflexive Design Studio to make our positions explicit, debatable, and open to critique. The roles and purposes we articulated are thus not final categories but openings—scaffolds for designers and researchers to interrogate, adapt, and expand through their own situated practices. In extending the framework to new contexts (e.g., educational studios, sustainability-oriented teams, or work with other living materials), new roles and purposes will likely emerge, enriching the vocabulary of MtH design and offering further ways of navigating the messy translation of theory into practice. In this way, our contributions offer a starting point: an invitation for others to test, contest, and expand the framework in diverse contexts, to surface new relationalities, and to advance MtH design as a practice-oriented inquiry. By doing so, we hope to contribute toward bridging MtH theories with their practical enactments and to foster more robust, transparent, and collective applied MtH design practices.

Acknowledgments

This research was funded by the Research Council of Finland's Flagship "Forest-Human-Machine Interplay -Building Resilience, Redefining Value Networks and Enabling Meaningful Experiences" (UNITE, 337653) and the Academy Research Fellowship project, PLAY-BIO (363282). We would also like to thank both the reviewers of this version and those of earlier submissions, whose comments have greatly strengthened our work.

References

[1] Ferran Altarriba Bertran, Oğuz'Oz' Buruk, Jordi Márquez Puig, and Juho Hamari. 2025. How Can Interactive Technology Help Us to Experience Joy With (in) the Forest? Towards a Taxonomy of Tech for Joyful Human-Forest Interactions. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. 1–21.

[2] Ferran Altarriba Bertran, Oğuz'Oz Buruk, Velvet Spors, and Juho Hamari. 2023. Playful inspiration for a new wave of joyful forest technology. In *Proceedings of the 2023 ACM Designing Interactive Systems Conference*. 1886–1903.

[3] Aphrodite Theodora Andreou, Stefano Gualeni, and Guillermo Jesús Sánchez Contreras. 2023. Co-designing enrichment toys with Bottlenose dolphins: playfulness as a corrective to anthropocentrism. In *Proceedings of the Tenth International Conference on Animal-Computer Interaction*. 1–12.

[4] Karen Barad. 2007. Meeting the universe halfway: Quantum physics and the entanglement of matter and meaning. *Duke Up* (2007).

[5] Armi Behzad, Ron Wakkary, Doenja Oogjes, Ce Zhong, and Henry Lin. 2022. Iterating through Feeling-with Nonhuman Things: Exploring repertoires for design iteration in more-than-human design. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts*. 1–6.

[6] Fiona Bell, Joshua Coffie, and Mirela Alistar. 2024. Bio-Digital Calendar: Attuning to Nonhuman Temporalities for Multispecies Understanding. In *Proceedings of the Eighteenth International Conference on Tangible, Embedded, and Embodied Interaction*. 1–15.

[7] Fiona Bell, Michelle Ramsahoye, Joshua Coffie, Julia Tung, and Mirela Alistar. 2023. μMe: Exploring the Human Microbiome as an Intimate Material for Living Interfaces. In *Proceedings of the 2023 ACM Designing Interactive Systems Conference*. 2019–2033.

[8] Eli Blevis, Youn Kyung Lim, Erik Stolterman, and Kevin Makice. 2008. The iterative design of a virtual design studio. *Techtrends: A Journal of the Association for Educational Communications and Technology* 52, 1 (2008), 74–83.

[9] Dean Brown, Liliana Ovalle, and Naho Matsuda. 2023. A Practice-Based Approach to Post-Human Computer Interaction: Design Notes from Nature Scenes. In *Proceedings of the Seventeenth International Conference on Tangible, Embedded, and Embodied Interaction*. 1–14.

[10] Laura D Cosio, Saurav Sarkar, Salik Tariq, Oğuz'Oz' Buruk, and Juho Hamari. 2024. Pine Cone Prowl: Embodying a Flying Squirrel in Forest. In *Proceedings of the 27th International Academic Mindtrek Conference*. 220–224.

[11] Aykut Coskun, Nazli Cila, Iohanna Nicenboim, Christopher Frauenberger, Ron Wakkary, Marc Hassenzahl, Clara Mancini, Elisa Giaccardi, and Laura Forlano. 2022. More-than-human Concepts, Methodologies, and Practices in HCI. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts*. 1–5.

[12] Anthony Cuthbertson. 2024. https://www.independent.co.uk/tech/robot-mushroom-biohybrid-robotics-cornell-b2606970.html

[13] Maria Puig de La Bellacasa. 2017. *Matters of care: Speculative ethics in more than human worlds*. Vol. 41. U of Minnesota Press.

[14] Amelia DeFalco. 2024. From Mushroom Men to Mycorrhizal Relations: Imagining Posthuman Aging and Care. In *The Palgrave Handbook of Literature and Aging*. Springer, 197–214.

[15] Gilles Deleuze, Félix Guattari, et al. 1987. A thousand plateaus.

[16] David Demeritt. 2005. Sarah Whatmore hybrid geographies organiser: Noel Castree hybrid geographies, relational ontologies and situated knowledges. *Antipode* 37, 4 (2005), 818–823.

[17] Jacques Derrida. 2020. *Deconstruction in a nutshell: A conversation with Jacques Derrida, with a new introduction*. Fordham University Press.

[18] Markéta Dolejšová, Danielle Wilde, Ferran Altarriba Bertran, and Hilary Davis. 2020. Disrupting (more-than-) human-food interaction: experimental design, tangibles and food-tech futures. In *proceedings of the 2020 ACM designing interactive systems conference*. 993–1004.

[19] Judith Dörrenbächer, Diana Löffler, and Marc Hassenzahl. 2020. Becoming a robot-overcoming anthropomorphism with techno-mimesis. In *Proceedings of the 2020 CHI conference on human factors in computing systems*. 1–12.

[20] Julia Fink. 2012. Anthropomorphism and human likeness in the design of robots and human-robot interaction. In *Social Robotics: 4th International Conference, ICSR 2012, Chengdu, China, October 29-31, 2012. Proceedings 4*. Springer, 199–208.

[21] Joshua Finzi. 2024. Fungal Rhizomes: Mycorrhizae and Posthumanism. *Green Letters* (2024), 1–16.

[22] Christopher Frauenberger. 2019. Entanglement HCI the next wave? *ACM Transactions on Computer-Human Interaction (TOCHI)* 27, 1 (2019), 1–27.

[23] Çağlar Genç, Ferran Altarriba Bertran, Linas Gabrielaitis, Esthiak Ahmed, and Velvet Spors. 2024. Designing Our Way Through Abstractions: Calling for More Practice-based More-than-Human Design Research. In *Proceedings of the Halfway to the Future Symposium*. 1–10.

[24] Çağlar Genç, Ferran Bertran, Linas Gabrielaitis, Eshtiak Ahmed, and Velvet Spors. 2024. Designing Our Way Through Abstractions: Calling for More Practice-based More-than-Human Design Research. In *Halfway to the Future (HTTF '24)*.

[25] David J Gunkel. 2018. The relational turn: third wave HCI and phenomenology. *New Directions in Third Wave Human-Computer Interaction: Volume 1-Technologies* (2018), 11–24.

[26] Lone Koefoed Hansen and Margrete Lodahl Rolighed. 2024. Encountering Human-Plant Relations: a Discussion of How Interaction Design Can encourage Human Sensibility to More-than-human Plants. In *Proceedings of the 2024 ACM Designing Interactive Systems Conference*. 1400–1411.

[27] Donna Haraway. 2013. A cyborg manifesto: Science, technology, and socialist-feminism in the late twentieth century. In *The transgender studies reader*. Routledge, 103–118.

[28] Donna Haraway. 2013. Situated knowledges: The science question in feminism and the privilege of partial perspective 1. In *Women, science, and technology*. Routledge, 455–472.

[29] Donna Haraway. 2018. Staying with the trouble for multispecies environmental justice. *Dialogues in Human Geography* 8, 1 (2018), 102–105.

[30] Donna Jeanne Haraway. 2003. The companion species manifesto: dogs, people, and significant otherness.

[31] Yuta Ikeya, Bahareh Barati, and Stephan Wensveen. 2025. Aesthetics in Designing with the Living: A Systematic Review of Critical Perspectives and Artefacts. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. 1–33.

[32] Emily Kennedy, Daphne Fecheyr-Lippens, Bor-Kai Hsiung, Peter H Niewiarowski, and Matthew Kolodziej. 2015. Biomimicry: A path to sustainable innovation. *Design Issues* 31, 3 (2015), 66–73.

[33] Rebecca Kleinberger, Jennifer Cunha, Megha M Vemuri, and Ilyena Hirskyj-Douglas. 2023. Birds of a Feather Video-Flock Together: Design and Evaluation of an Agency-Based Parrot-to-Parrot Video-Calling System for Interspecies Ethical Enrichment.. In *Proceedings of the 2023 CHI conference on human factors in computing systems*. 1–16.

[34] Madlen Kneile, Judith Dörrenbächer, Marc Hassenzahl, and Matthias Laschke. 2025. Mediating Human-Nature Relations Through Technology. A Scoping Review of Post-Anthropocentric Artifacts and Their Design Strategies. *ACM Transactions on Computer-Human Interaction* (2025).

[35] Stacey Kuznetsov, Cassandra Barrett, Piyum Fernando, and Kat Fowler. 2018. Antibiotic-responsive bioart: Exploring DIYbio as a design studio practice. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*. 1–14.

[36] Zhenchi Lai, Janet Yi-Ching Huang, and Rung-Huei Liang. 2024. AI Cat Narrator: Designing an AI Tool for Exploring the Shared World and Social Connection with a Cat. In *Companion Publication of the 2024 ACM Designing Interactive Systems Conference*. 224–228.

[37] Shaun Lawson, Ben Kirman, and Conor Linehan. 2016. Power, participation, and the dog internet. *Interactions* 23, 4 (2016), 37–41.

[38] Jen Liu, Daragh Byrne, and Laura Devendorf. 2018. Design for collaborative survival: An inquiry into human-fungi relationships. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*. 1–13.

[39] Henry Lowe, Ngeh Toyang, Blair Steele, Henkel Valentine, Justin Grant, Amza Ali, Wilfred Ngwa, and Lorenzo Gordon. 2021. The therapeutic potential of psilocybin. *Molecules* 26, 10 (2021), 2948.

[40] Andrés Lucero. 2015. Using affinity diagrams to evaluate interactive prototypes. In *Human-Computer Interaction–INTERACT 2015: 15th IFIP TC 13 International Conference, Bamberg, Germany, September 14-18, 2015, Proceedings, Part II 15*. Springer, 231–248.

[41] Tatu Marttila. 2011. Unpleasurable products and interfaces: provocative design communication for sustainable society. In *Proceedings of the 2011 conference on Designing Pleasurable Products and Interfaces*. 1–4.

[42] Livia J Müller, Elisa D Mekler, and Klaus Opwis. 2015. Facets in HCI: towards understanding eudaimonic UX–preliminary findings. In *Proceedings of the 33rd Annual ACM Conference Extended Abstracts on Human Factors in Computing Systems*. 2283–2288.

[43] Iohanna Nicenboim, Joseph Lindley, Cristina Zaga, Arne Berger, Laura Forlano, and Elisa Giaccardi. 2024. More-Than-Human Design in Practice. In *DRS2024*, Ciliotta Chehade E. Hekkert P. Forlano L. Ciuccarelli P. Lloyd P. Gray, C. (Ed.).

[44] Iohanna Nicenboim, Doenja Oogjes, Heidi Biggs, and Seowoo Nam. 2024. Decentering Through Design: Bridging Posthuman Theory with More-than-Human Design Practices. *Human–Computer Interaction* (2024), 1–26.

[45] Netta Ofer and Mirela Alistar. 2024. Tracing as a Strategy for Orienting to Nonhuman Perspectives. In *Proceedings of the 2024 ACM Designing Interactive Systems Conference*. 1087–1100.

[46] Gizem Oktay, Minha Lee, Bahareh Barati, and Ron Wakkary. 2024. Caring through: Engaging with temporality of care in more-than-human design. In *Proceedings of the Halfway to the Future Symposium*. 1–8.

[47] Patricia Pons, Javier Jaen, and Alejandro Catala. 2015. Envisioning future playful interactive environments for animals. *More playful user interfaces: Interfaces that invite social and physical interaction* (2015), 121–150.

[48] Julia Reade. 2022. "What Would the Mushrooms Say?" Speculating Inclusive and Optimistic Futures with Nature as Teacher. *Humanities* 11, 4 (2022), 97.

[49] Irene Roncero-Ramos and Cristina Delgado-Andrade. 2017. The beneficial role of edible mushrooms in human health. *Current Opinion in Food Science* 14 (2017), 122–128.

[50] Anton Poikolainen Rosén, Maria Normark, and Mikael Wiberg. 2022. Noticing the environment–A design ethnography of urban farming. In *Nordic Human-Computer Interaction Conference*. 1–13.

[51] Chris Rust. 2004. Design enquiry: Tacit knowledge and invention in science. *Design issues* 20, 4 (2004), 76–85.

[52] Erik Sandelin and Sarah Homewood. 2020. Design (in) actions. In *Proceedings of the 11th Nordic Conference on Human-Computer Interaction: Shaping Experiences, Shaping Society*. 1–9.

[53] Donald A Schön. 1985. *The design studio: An exploration of its traditions and potentials*. RIBA Publications for RIBA Building Industry Trust London.

[54] Marie Louise Juul Søndergaard and Nadia Campo Woytuk. 2023. Feminist posthumanist design of menstrual care for more-than-human bodies. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–18.

[55] Brian Sullivan. 2015. *The design studio method: creative problem solving with UX sketching*. Routledge.

[56] Oscar Tomico, Doenja Oogjes, and Ron Wakkary. 2025. Constituency as a Matter of Practice: Moving a Plant Studio. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. 1–17.

[57] Oscar Tomico, Anton Poikolainen Rosén, Svenja Keune, Ferran Altarriba Bertran, Danielle Wilde, Daniel Fernández Galeote, Tau Ulv Lenskjold, Ruut Tikkanen, Oğuz'Oz Buruk, and Velvet Spors. 2024. Seeding a repository of methods-to-be for nature-entangled design research. In *Proceedings of the 2024 ACM Designing Interactive Systems Conference*. 1101–1115.

[58] Oscar Tomico, Ron Wakkary, and Kristina Andersen. 2023. Living-with and Designing-with Plants. *Interactions* 30, 1 (2023), 30–34.

[59] Anna Tsing. 2012. Unruly edges: mushrooms as companion species: for Donna Haraway. *Environmental humanities* 1, 1 (2012), 141–154.

[60] Anna Tsing. 2015. The Mushroom at the End of the World: On the Possibility of Life in Capitalist Ruins.

[61] Ron Wakkary. 2021. *Things we could design: For more than human-centered worlds*. MIT press. https://doi.org/10.7551/mitpress/13649.001.0001

[62] Ron Wakkary, Doenja Oogjes, Nazmus Sakib, and Armi Behzad. 2023. Turner Boxes and Bees: From Ambivalence to Diffraction. In *Proceedings of the 2023 ACM Designing Interactive Systems Conference*. 790–807.

[63] Ron Wakkary, Doenja Oogjes, Oscar Tomico, Nazmus Sakib, and Ege Kökel. 2025. Backyard Practices: A Liminal Approach to Designing in More-than-Human Worlds. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. 1–18.

[64] John Zimmerman, Jodi Forlizzi, and Shelley Evenson. 2007. Research through design as a method for interaction design research in HCI. In *Proceedings of the SIGCHI conference on Human factors in computing systems*. 493–502.